2.6 Geometry and Locus of Complex Numbers

In this section let us study the geometrical interpretation of complex number z in complex plane and the locus of z in Cartesian form.

Example 2.18

Given the complex number $z = 3 + 2i$, represent the complex numbers z, iz, and $z + iz$ in one Argand plane. Show that these complex numbers form the vertices of an isosceles right triangle.

Solution

Given that $z = 3 + 2i$.

Therefore, $iz = i(3 + 2i) = -2 + 3i$

$z + iz = (3 + 2i) + i(3 + 2i) = 1 + 5i$

Let A, B, and C be z, $z + iz$, and iz respectively.

$AB^2 = |(z + iz) - z|^2 = |-2 + 3i|^2 = 13$

$BC^2 = |iz - (z + iz)|^2 = |-3 - 2i|^2 = 13$

$CA^2 = |z - iz|^2 = |5 - i|^2 = 26$

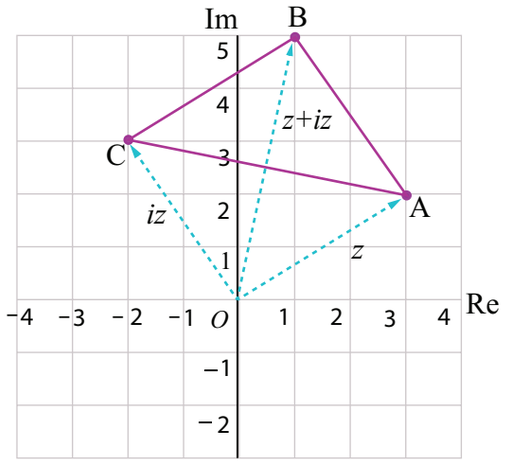


Fig. 2.22

Since $AB^2 + BC^2 = CA^2$ and $AB = BC$, ΔABC is an isosceles right triangle.

Definition 2.5 (circle)

A circle is defined as the locus of a point which moves in a plane such that its distance from a fixed point in that plane is always a constant. The fixed point is the centre and the constant distant is the radius of the circle.

Equation of Complex Form of a Circle

The locus of z that satisfies the equation $|z - z_0| = r$ where z_0 is a fixed complex number and r is a fixed positive real number consists of all points z whose distance from z_0 is r.

Therefore $|z - z_0| = r$ is the complex form of the equation of a circle. (see Fig. 2.23)

(i) $|z - z_0| < r$ represents the points interior of the circle.

(ii) $|z - z_0| > r$ represents the points exterior of the circle.

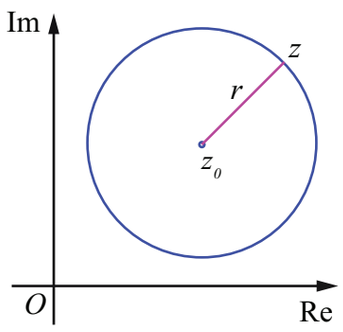


Fig. 2.23

Illustration 2.3

$|z| = r \Rightarrow \sqrt{x^2 + y^2} = r$

$\Rightarrow x^2 + y^2 = r^2$, represents a circle centre at the origin with radius r units.

12th_Maths_EM_Vol1_CH 2_Complex Numbers.indd 73 16-12-2022 12:27:59